



**SCHOOL OF COMPUTING AND INFORMATION SCIENCES
FACULTY OF MANAGEMENT AND INFORMATION SCIENCES**

Final Year Project (FYP) – Topic Approval Document

1. Administrative Details

Program	BSCS	Academic Year	2026
Group ID	(To be Assigned)	Submission Date	April 23, 2026

2. Student / Group Information

Student Name	Registration No.	Role in Project
Umer Ahmed	BSCS-01011	AI Model + Development
Kashan M. Ashraf	BSCS-01007	Video Processing + Documentation
Imran	BSCS-01014	UI Design + Testing

3. Proposed Project Information

Project Title

Real-Time Automated Fall Detection and Alert System for Elderly and Patient Safety in Low-Resource Environments

Problem Statement

In Pakistan, elderly care and patient monitoring are severely limited due to staff shortages and the absence of smart healthcare infrastructure. Falls in homes and hospitals frequently go unnoticed, leading to serious and often life-threatening injuries. Existing solutions are either too costly, require wearable sensors, or rely on passive CCTV with no intelligent detection capability. This project proposes an affordable AI-based system that detects falls in real time using existing camera feeds, processes video locally, and immediately alerts caregivers making it practical for homes, clinics, and low-resource healthcare settings across Pakistan.

4. Real-World Context

Where Does This Problem Occur?

This problem is prevalent in hospitals, small clinics, and healthcare centers across Karachi and other Pakistani cities, as well as in homes where elderly individuals live independently or with limited supervision. The issue is especially critical in rural areas where professional medical supervision is scarce or inaccessible.

Who Faces This Problem?

The primary stakeholders include:

- Elderly individuals living alone or with limited caregiver support
- Patients recovering in hospitals or home care settings
- Caregivers and family members responsible for patient safety
- Hospital staff who cannot monitor all patients simultaneously

How Is This Problem Currently Handled?

Currently, the problem is managed through manual monitoring by caregivers or hospital staff, and through passive CCTV surveillance systems that lack any intelligent detection capabilities. These approaches are reactive, labor-intensive, prone to human error, and often fail to respond in time to prevent injury or complications following a fall.

5. Motivation

Falls among the elderly and bedridden patients represent one of the most common and preventable causes of serious injury in Pakistan. Manual monitoring is unsustainable given the caregiver-to-patient ratio in most local healthcare settings, and advanced commercial smart systems are prohibitively expensive. This project addresses a real and urgent need by proposing a low-cost, camera-based AI solution that leverages existing CCTV or webcam infrastructure. By enabling real-time, automated detection and immediate alert generation, the system aims to reduce response time, minimize injury severity, and lower the burden on caregivers delivering meaningful, measurable impact to an underserved problem in the Pakistani healthcare context.

6. Scope Definition

In Scope (What the Project WILL Do)

- Real-time fall detection using live camera feed (webcam or CCTV)
- Support for offline testing using pre-recorded video files
- Rule-based fall detection using body pose orientation and movement thresholds (FYP-1)
- Machine learning and deep learning-based detection using LSTM model (FYP-2)
- Automated alert system with audible sound and on-screen notification upon fall detection
- Basic UI dashboard displaying live camera feed and current detection status
- Event logging system recording timestamped fall events for review

- Support for challenging scenarios: low-light conditions, partial occlusion, and local furniture (e.g., charpai)

Out of Scope (What the Project Will NOT Do)

- Medical diagnosis, prognosis, or any health condition prediction
- Cloud-based or large-scale enterprise-level deployment
- Continuous 24/7 hospital-grade monitoring infrastructure
- Integration with Electronic Health Record (EHR) systems
- Multi-camera tracking across multiple rooms simultaneously

7. Proposed Solution Overview

High-Level Solution Description

The proposed system is an AI-based fall detection solution that uses camera input to monitor human activities in real time. It processes live or recorded video, detects persons in frame, estimates body pose through key point extraction, and analyzes movement patterns to identify falls. On detection, the system generates an immediate alert and logs the event. The system is designed to be deployable on standard consumer hardware without any specialized infrastructure.

System Definition

Inputs:

- Live camera feed (webcam or CCTV)
- Recorded video files (.mp4, .avi)

Processing Modules:

Module No.	Module Name	Function
1	Video Capture Module	Reads frames from live camera or video file
2	Person Detection Module	Identifies and locates persons in each frame using YOLOv8
3	Pose Estimation Module	Extracts body keypoints (skeleton) using MediaPipe
4	Fall Detection Engine	Analyzes posture and motion to classify: Normal / Warning / Fall
5	Alert Generation Module	Triggers audible alarm and on-screen notification upon fall
6	Logging & Monitoring Module	Records timestamped fall events and displays history in dashboard

System Workflow:

Camera Input → Person Detection → Pose Extraction → Motion Analysis → Fall Classification → Alert Trigger → Log Storage → Dashboard Display

Outputs:

- Fall / No Fall classification per frame
- Real-time alert (sound + on-screen notification)
- Timestamped event log of all fall detections
- Live dashboard with camera feed and system status

Users:

- Caregivers and family members monitoring elderly at home
- Hospital or clinic staff overseeing patient wards

Key Functional Features

1. Real-time human pose detection from live or recorded camera input
2. Automated fall detection based on body posture orientation and movement analysis
3. Three-class activity classification: Normal, Warning (potential fall), Fall detected
4. Immediate alert generation (audio alarm + on-screen notification) upon fall detection
5. Event logging with timestamps and a dashboard for live feed and fall history review

8. Existing Solutions & Differentiation

Is This Problem Already Addressed?

☒ Yes

Comparison of Existing Solutions vs. Proposed System

Existing Solution	Key Limitations	How Our System Improves
CCTV Surveillance Systems	No intelligent detection; requires constant human monitoring	Adds AI-based automated detection and real-time alerts without replacing existing cameras
Wearable Fall Detection Devices (smart bands/watches)	Requires device purchase and consistent wearing; uncomfortable for patients; battery-dependent	Camera-based; no wearable needed; works passively without patient cooperation
Manual Monitoring by Caregivers	Labor-intensive; prone to human error; not feasible 24/7	Automates monitoring continuously, reducing caregiver load and eliminating attention gaps

Proposed Differentiation Summary

Unlike existing approaches, this system uses a low-cost, camera-based AI pipeline that works with already-installed cameras without requiring wearable devices or dedicated hardware. It provides real-time, pose-based fall detection with automatic alerting, and is designed specifically for the resource-constrained home and small clinic environments typical in Pakistan. The system is deployable on standard consumer PCs (Intel i5, 8GB RAM) and operates with no cloud dependency.

9. Technical & Design Considerations

Proposed System Architecture

The system follows a sequential AI-based computer vision pipeline structured in six layers:

Layer	Description
Input Layer	Accepts live webcam/CCTV feed or pre-recorded video files via OpenCV VideoCapture
Detection Layer	YOLOv8 identifies and localizes persons in each video frame in real time
Pose Estimation Layer	MediaPipe extracts around 33 body key points (skeleton landmarks) per detected person
Analysis Layer	Rule-based engine (FYP-1) / LSTM model (FYP-2) analyzes key point angles, vertical displacement, and temporal motion sequences to classify activity state
Alert & Output Layer	Triggers audio alarm and on-screen notification; outputs Fall/No Fall classification
Logging & Dashboard Layer	Records events with timestamps; displays live feed, detection status, and fall history

Tools & Technologies

Tool / Library	Version / Type	Purpose
MediaPipe	Google (Python)	Real-time human pose estimation and key point extraction
Ultralytics YOLOv8	Deep Learning Model	Person detection and bounding box localization
OpenCV	Python Library	Video capture, frame preprocessing, display rendering

Tool / Library	Version / Type	Purpose
TensorFlow / Keras	ML Framework	LSTM model training for sequential fall classification (FYP-2)
Python (Tkinter / PyQt)	GUI Framework	Basic UI dashboard development

Major Technical Decisions

Decision	Justification
MediaPipe + YOLOv8 for Detection & Pose	Lightweight, real-time capable on standard hardware (Intel i5, 8GB RAM). MediaPipe provides around 33 key points with low latency; YOLOv8 provides robust person localization.
Rule-Based Detection in FYP-1	Ensures a stable, testable baseline before applying ML. Logic is based on body orientation angles and center-of-mass displacement interpretable and debuggable.
LSTM Model in FYP-2	Falls are temporal events. LSTM captures sequential motion patterns across frames, reducing false positives from sudden non-fall movements (e.g., sitting quickly).
Local Processing (No Cloud)	Eliminates network dependency, reduces latency, and protects user privacy critical for healthcare settings.

Handling Environmental Challenges

Occlusion Handling:

To handle situations where furniture (e.g., chairs, beds, charpai) partially obstructs a person during a fall, the system will apply temporal analysis by tracking body key points across multiple consecutive frames rather than relying on any single static frame. This frame-sequence approach maintains state continuity even during momentary occlusions.

Lighting Variability:

To ensure reliability in low-light conditions (e.g., night-time in hospitals or poorly lit homes), the system will apply adaptive contrast enhancement and pre-processing filters within the OpenCV pipeline. These techniques normalize input frames so the pose estimation model receives consistent, high-contrast data regardless of ambient lighting.

10. Dataset Strategy

Datasets to Be Used

Dataset	Details
UR Fall Detection Dataset	Public benchmark dataset with labelled fall and non-fall video sequences from camera
Multiple Cameras Fall Dataset (MCFD)	Multi-angle fall event recordings for improved generalization
Self-Recorded Data	Custom videos recorded by team members simulating falls in local settings (home-like environments, charpai, low-light) to address dataset bias toward non-Pakistani contexts

Data Split Strategy

Split	Proportion	Purpose
Training	~70%	Model learning
Validation	~15%	Hyperparameter tuning and early stopping
Testing	~15%	Final performance evaluation on unseen data

11. Measurable Evaluation Criteria

Performance Metrics

Metric	Definition	Target
Accuracy	Proportion of correctly classified events (fall / no fall)	$\geq 90\%$
Precision	Of all detected falls, proportion that were actual falls	$\geq 85\%$
Recall (Sensitivity)	Of all actual falls, proportion correctly detected	$\geq 90\%$
False Positive Rate	Rate of non-fall events incorrectly classified as falls	$\leq 10\%$

Metric	Definition	Target
Alert Response Time	Time elapsed between fall event and alert generation	<= 2 seconds
Processing Speed (FPS)	Frames processed per second during live detection	>= 15 FPS

Test Scenarios

The system will be validated across the following specific test scenarios:

- Standard fall on a flat floor in good lighting
- Fall partially obscured by furniture (chair, bed, charpai)
- Detection in low-light or night-time conditions
- Sudden non-fall movements (sitting down rapidly, bending) false positive resistance
- Multiple persons in frame simultaneously
- Recorded video file testing for accuracy benchmarking

12. Feasibility & Risk Assessment

Key Risks & Mitigation Plan

#	Risk	Category	Mitigation Strategy
1	False alarms (non-fall classified as fall)	Detection Accuracy	Implement multi-frame confirmation logic; require fall state to persist across N consecutive frames before triggering alert
2	Missed detections (fall not detected)	Detection Accuracy	Combine pose estimation with YOLOv8 bounding box tracking; temporal LSTM model in FYP-2 to capture slow or gradual falls
3	Poor performance in low-light environments	Environmental	Apply OpenCV-based adaptive contrast enhancement and frame normalization in preprocessing stage
4	Occlusion by local furniture (charpai, chair)	Environmental	Temporal multi-frame tracking to maintain pose state continuity during momentary occlusion
5	Real-time processing constraints on low-end hardware	Technical	Optimize pipeline with frame skipping and model pruning; validated on Intel i5 with 8GB RAM

#	Risk	Category	Mitigation Strategy
6	Dataset bias (non-Pakistani fall patterns)	Data	Supplement public datasets with self-recorded videos in local environments to improve model generalization
7	Limited public dataset size for training	Data	Apply data augmentation (rotation, brightness, flip); use transfer learning with pre-trained pose models
8	Skill gap in deep learning (FYP-2)	Team	Divide tasks by strength; begin FYP-2 ML work in parallel with rule-based prototype; use well-documented TensorFlow/Keras APIs

13. Expected Deliverables

Deliverable	Details
☑ Working System / Prototype	Functional fall detection demo with live camera and alert system
☑ Project Documentation	Complete technical and user documentation
☑ Demonstration / Presentation	Live demo and presentation to supervisor/panel
☑ Testing & Validation Evidence	Accuracy reports, test scenarios, and benchmark results

14. Student Declaration

We confirm that:

- This project is based on a real-world problem
- We are aware of existing solutions and their limitations
- The idea and work presented in this proposal are our own
- We understand that copy-paste or cloned projects are not acceptable

Umer Ahmed

Signature

Kashan M. Ashraf

Signature

Imran

Signature

15. Faculty Evaluation (Official Use Only)

Evaluation Area	Satisfactory	Needs Revision	Unsatisfactory
Problem Clarity	☐	☐	☐
Real-World Relevance	☐	☐	☐
Scope Realism	☐	☐	☐
Awareness of Existing Solutions	☐	☐	☐
Differentiation Justification	☐	☐	☐
Engineering Depth	☐	☐	☐
Feasibility	☐	☐	☐
Ownership	☐	☐	☐

15. Decision & Remarks

Decision

- ☐ Working System / Prototype
- ☐ Approved with Revisions
- ☐ Not Approved

Required Revisions (if any)

Resubmission Deadline (if applicable)

15. Signatures

FYP Coordinator: _____ Date: _____

Supervisor: _____ Date: _____